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FieldNet: A LoRa-Powered IoT Framework for Agricultural Automation

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Motivation
Literature Survey
Methodology
Design
Hardware Implementation
Working Flow
Results
Conclusion
References

🌱 Motivation

- Need for real-time, reliable, and long-range agriculture monitoring
- Traditional methods are manual, time-consuming, and less accurate
- Existing wireless tech is either short-range, high-power, or expensive
- LoRa offers a solution
- ⚡ Low power
- 📡 Long range
- 💰 Cost-effective

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Literature Survey

1. LoRa-Based Smart Agriculture Monitoring and Automatic Irrigation System, Xiaoming Ding et al., 2024
Address inefficiencies in traditional agriculture, such as water waste and lack of real-time monitoring, by implementing a LoRa-based wireless system.
Developed a system with sensor nodes, gateways, and actuator nodes. Data is collected and transmitted via LoRa, visualised in real time using Blynk, and stored/analysed on ThingSpeak.

Key Findings
- Achieved reliable long-range communication (>6 km) with low power consumption.
- Enabled automated irrigation based on real-time sensor data and user commands.

2. Optimising Performances of LoRa-Based IoT Enabled Wireless Agriculture Science Direct, 2023
Optimise LoRa parameter settings for affordable, scalable monitoring on fruit and vegetable farms.
Experimented with LoRa configurations (spreading factor, bandwidth, transmission power) to maximise coverage and minimise energy use.

Key Findings
- Proper tuning of LoRa parameters can significantly extend battery life and coverage.
- Demonstrated the importance of balancing data rate and range for agricultural deployments.

3.An Efficient LoRa Smart Agriculture Platform Using WSNs JETIR, 2023
Build a LoRa-based smart agriculture management and monitoring system using wireless sensor networks (WSNs).
Created a private LoRa network interfaced with a gateway, collecting data from end nodes and transmitting it to the cloud for user applications.

Key Findings
- LoRa solves communication failure and energy-saving challenges in rural agricultural monitoring.
- Field tests confirmed satisfactory coverage and low power operation, even in dense environments.

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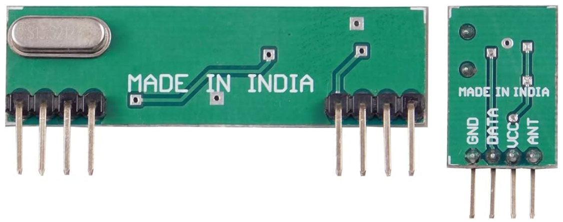
Previous Method

RF Module

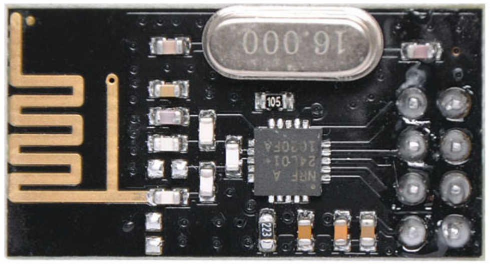


1. Easy For RF-based Application.
2. Complete Radio Transmitter.
3. Transmit Range Up To 50m.
4. CMOS / TTL Input.

RF zigbee Module

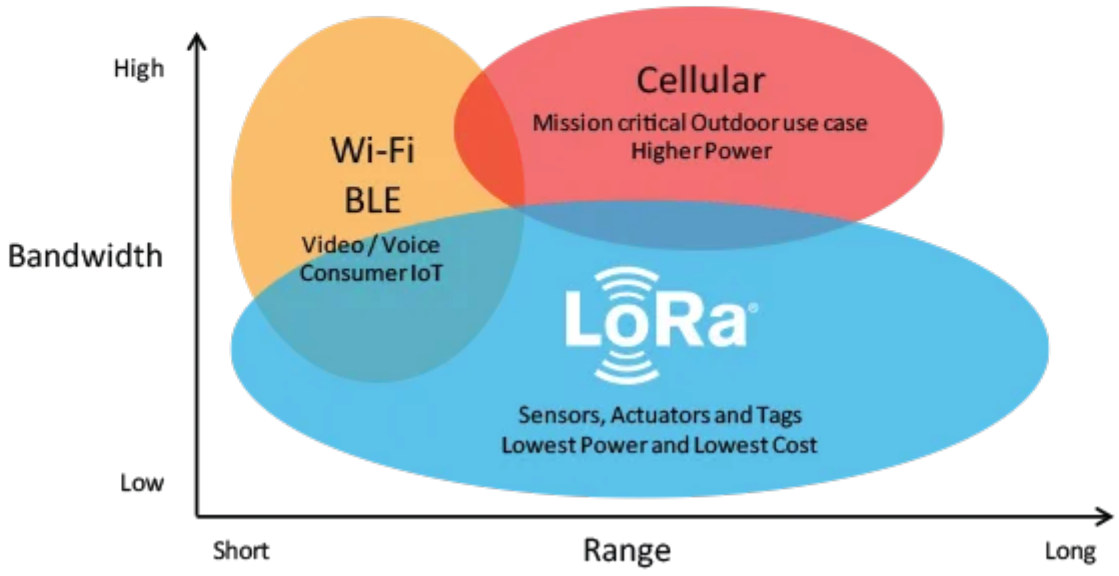
1. Integrated, Wire Antenna
2. 15 general-purpose I/O lines
3. Industry-leading sleep current of sub 1 μA
4. Transmit Range Up To 1km.

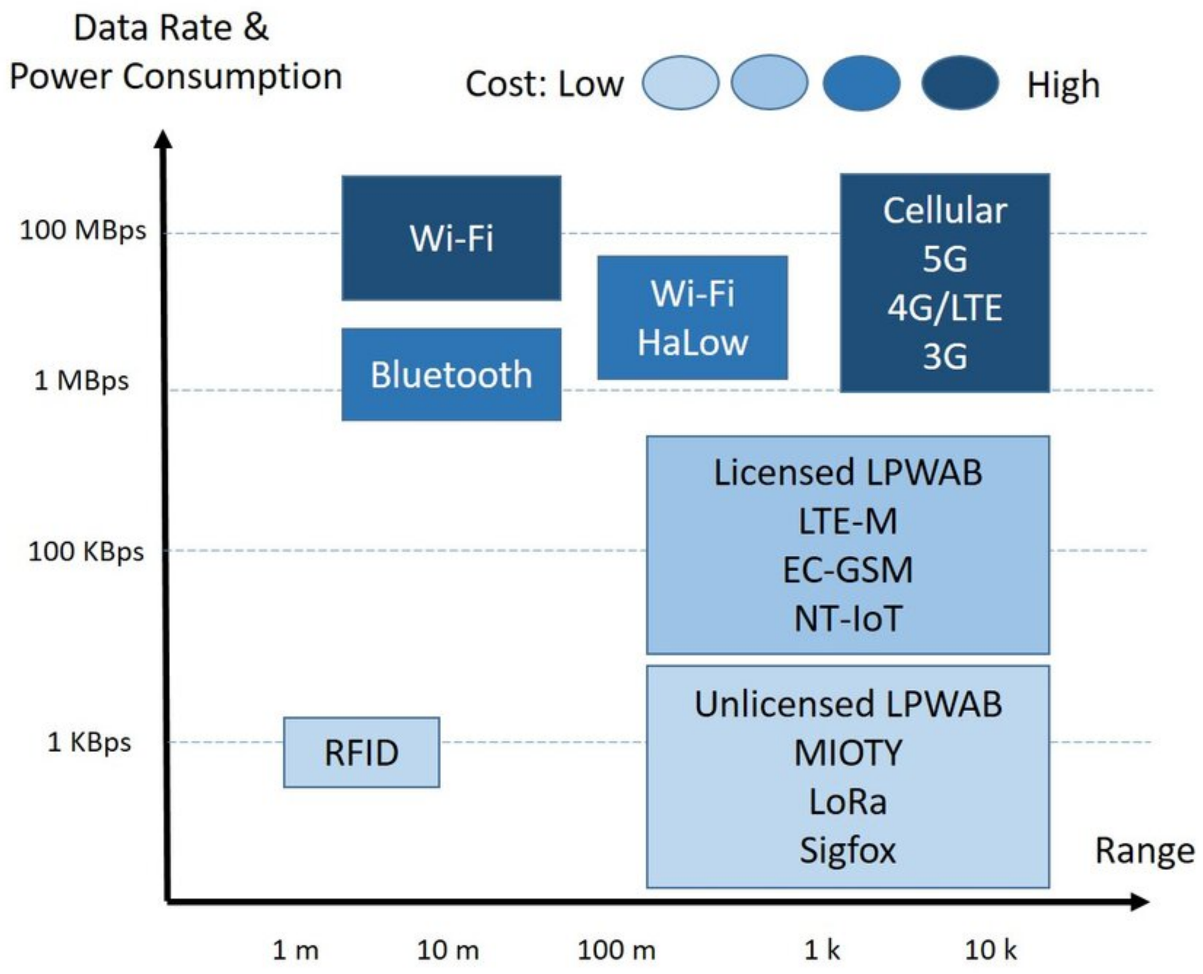
RF NRF Module



1. Power-down mode current: 4.2uA.
2. Operating Range: 1Km
3. Receive Mode Current(peak): 45mA
4. It uses 2.4GHz global open ISM band.

🛠️ LoRa & Other Wireless Technologies

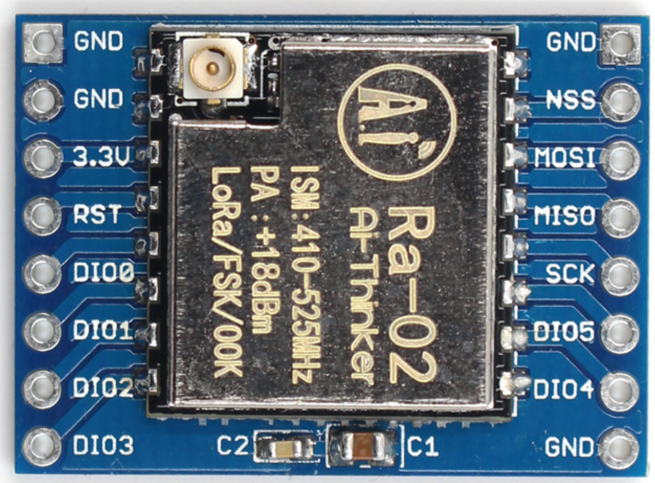


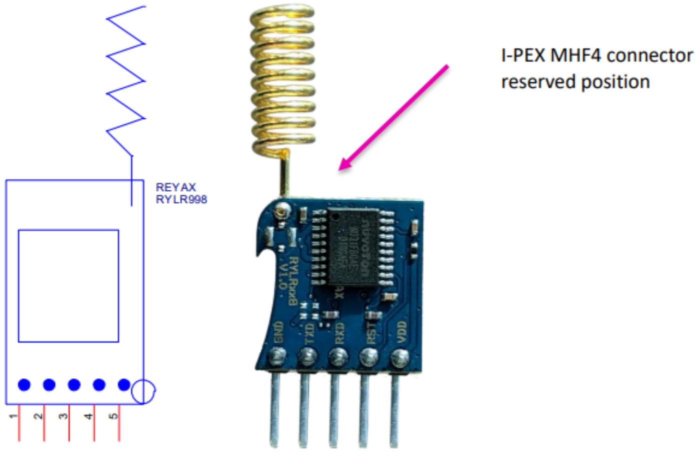


LoRa Data Rates = 0.3 kbps to 50 kbps (FSK)

SX126x Series = 1.2 μw to 10 μw

🧪 Our Method





Frequency Bands 169 MHz, 433 MHz, 868 MHz and 915 MHz

📡 Long Range
Transmits data reliably across several kilometers, covering large farms and remote fields

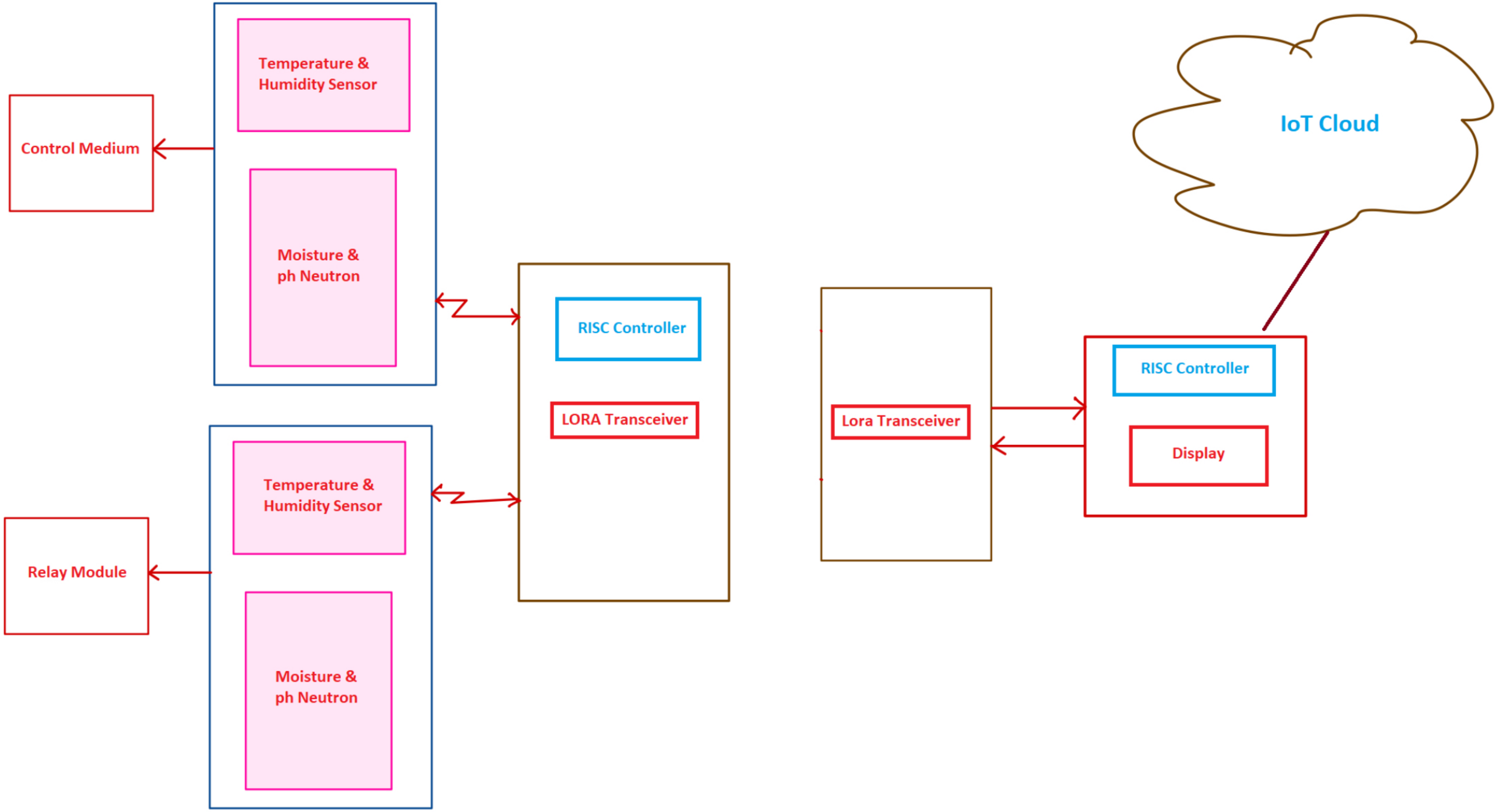
⚡ Low Power Consumption
Sensors can operate for years (up to 1 years in some cases)

💰 Cost-Effective
Lower setup and operational costs compared to cellular or other wireless solutions

🛡️ Robust and Reliable
Strong signal penetration through obstacles resistance to interference, and secure, encrypted data transmission.

🔗 Easy Integration & Scalability
supports many sensor nodes

⚡ Block Diagram



⚡ Block Diagram

1. 🧠 RISC Microcontroller
   Central processing unit that collects sensor data and manages communication.

2. 🌱 Soil Sensors (NPK, Moisture, Temperature)
   Measure key soil parameters to monitor nutrient levels and environmental conditions.

3. 📡 LoRa Module (SX1278)
   Enables long-range, low-power wireless transmission of sensor data to the gateway.

4. 🖥️ Displays (OLED)
   Show real-time sensor readings on-site for immediate monitoring.

5. ☁️ Cloud/Server & User Interface
   Stores, visualizes, and analyzes data remotely
   allows farmers to monitor and control irrigation via web or mobile apps.

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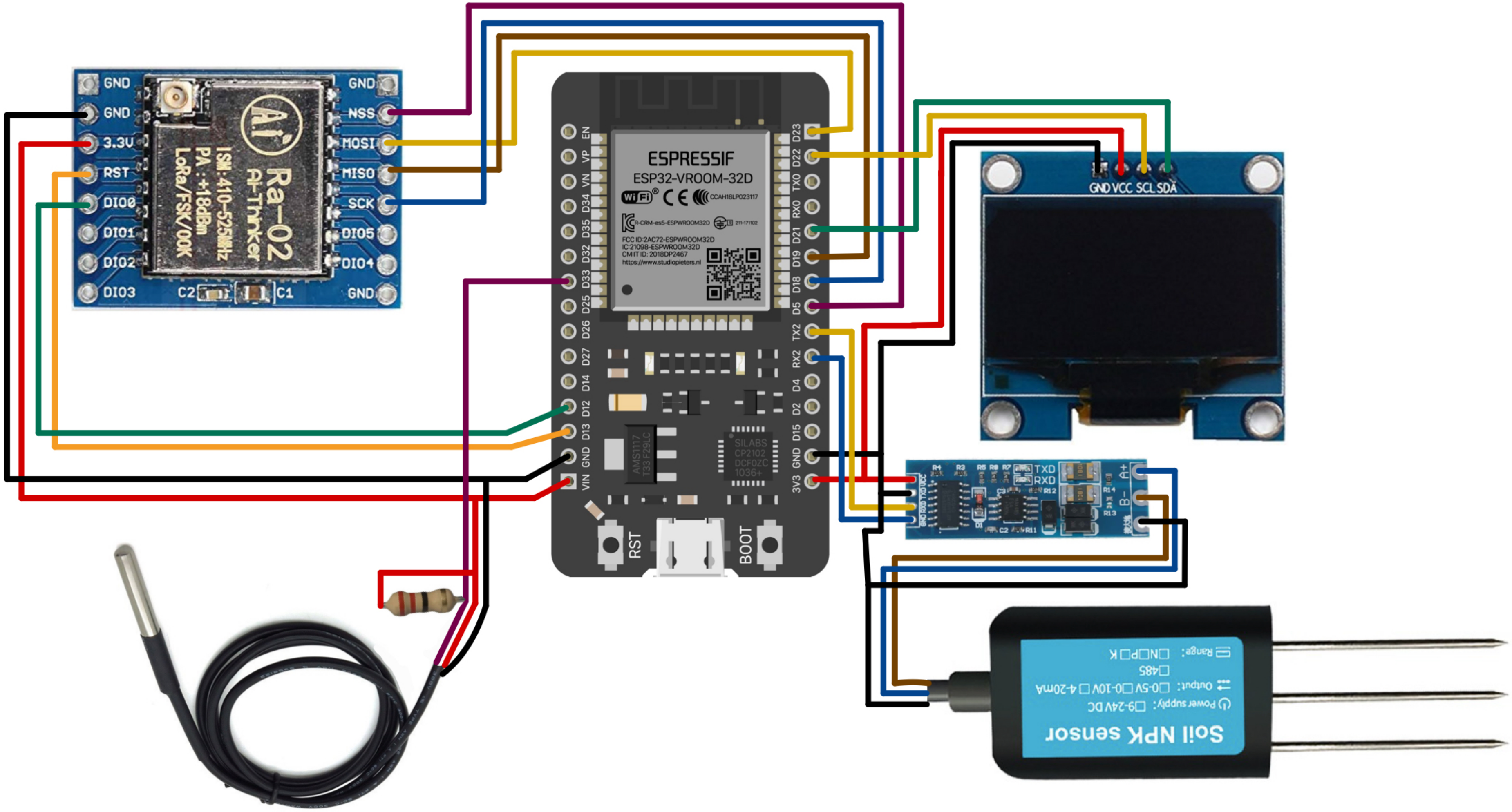
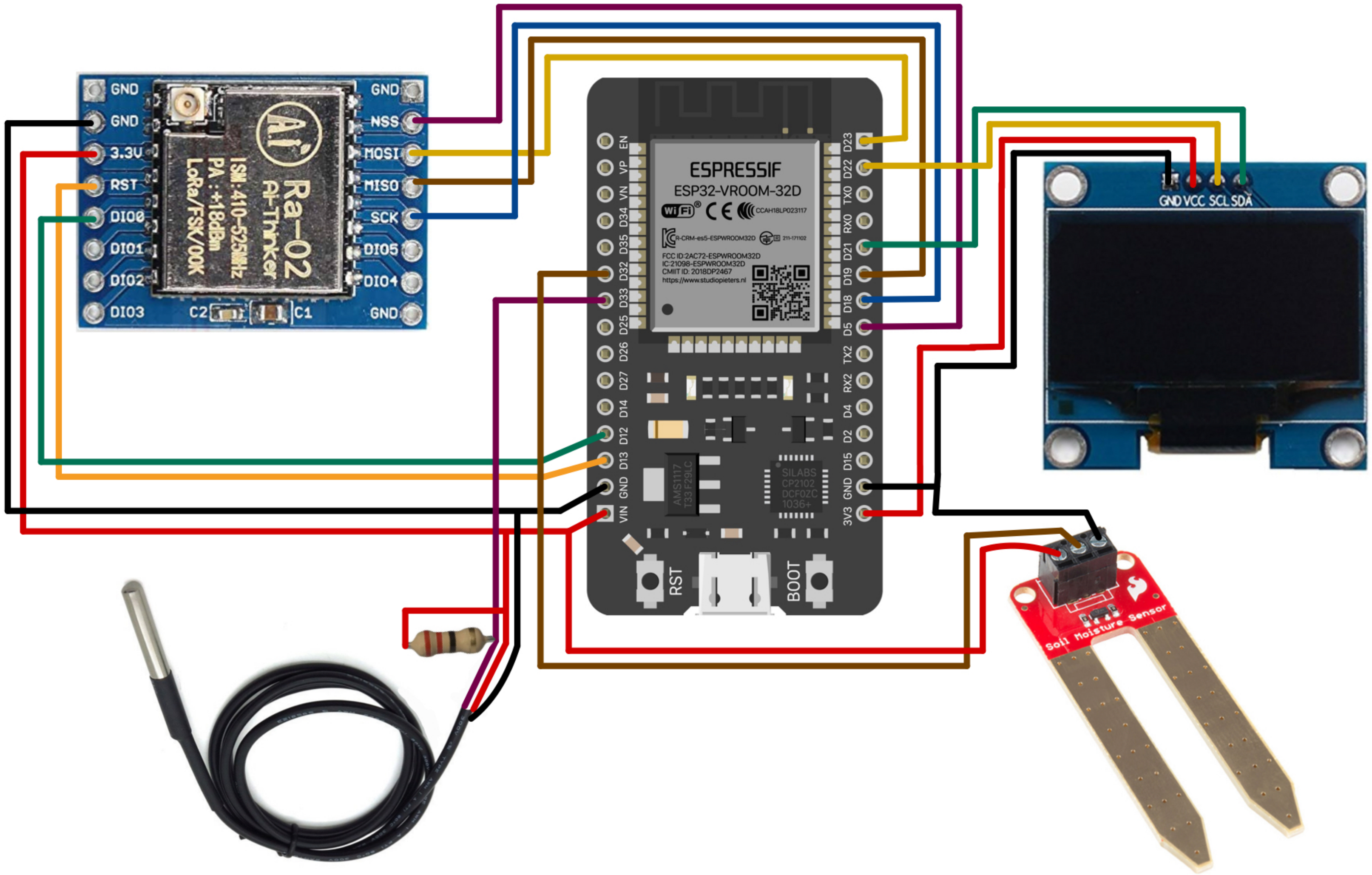
Node_1



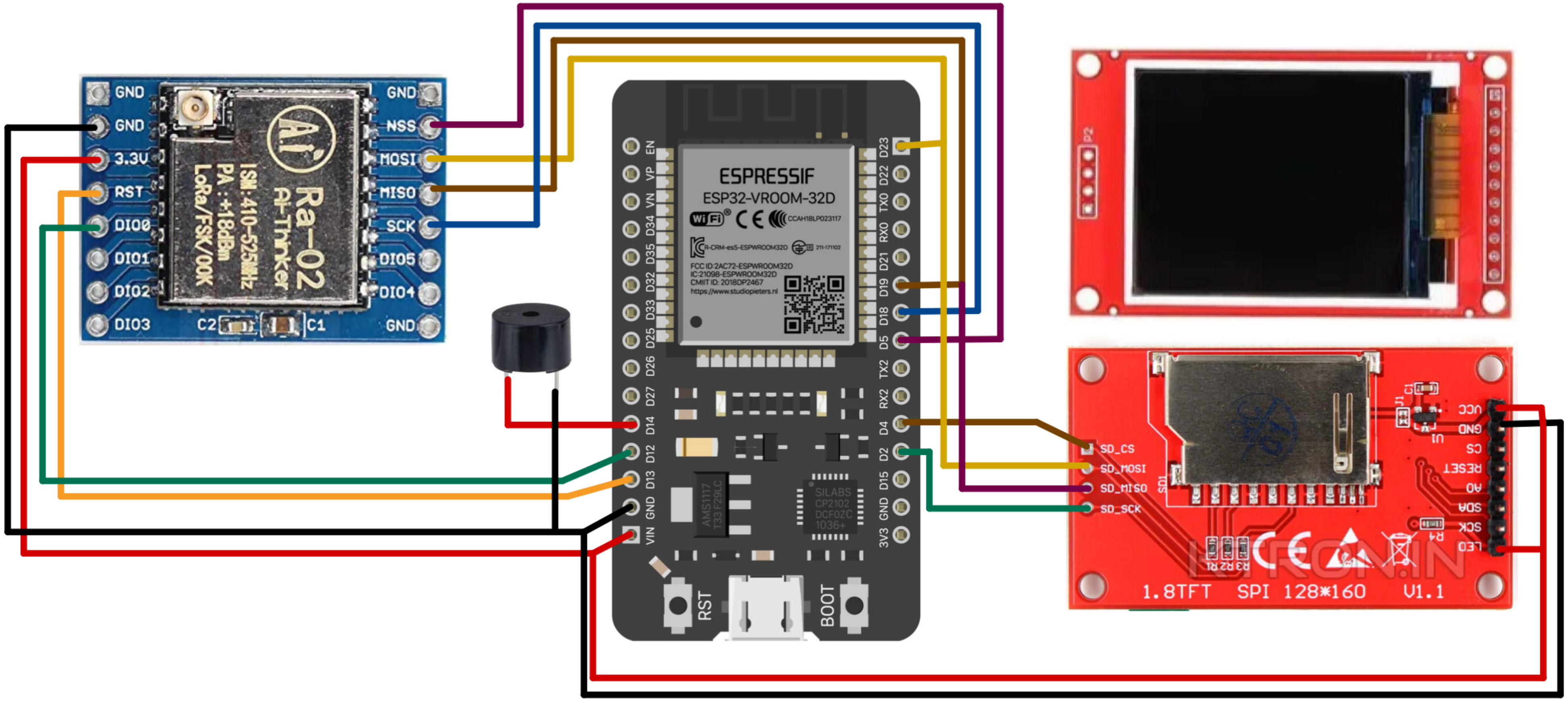
GND
GND
3.3V
RST
DIO0
DIO1
DIO2
DIO3
GND
NSS
MOSI
MISO
SCK
DIO5
DIO4
GND
Ra-02
Ai-Thinker
ESPRESSIF
ESP32-VROOM-32D
EN
VP
VN
D34
D35
D32
D33
D25
D26
D27
D14
D12
D13
GND
VIN
D23
D22
TX0
RX0
D21
D19
D18
D5
TX2
RX2
D4
D2
D15
GND
3V3
RST
BOOT
GND VCC SCL SDA
TXD
RXD
Soil NPK sensor

Node_2

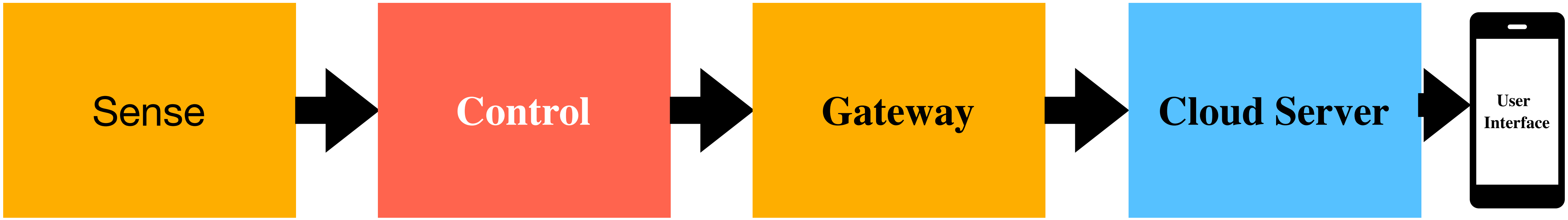


GND
GND
3.3V
RST
DIO0
DIO1
DIO2
DIO3
NSS
MOSI
MISO
SCK
DIO5
DIO4
Ra-02
Ai-Thinker
ESPRESSIF
ESP32-VROOM-32D
EN
VP
VN
D34
D35
D32
D33
D25
D26
D27
D14
D12
D13
GND
VIN
D23
D22
TX0
RX0
D21
D19
D18
D5
TX2
RX2
D4
D2
D15
GND
3V3
RST
BOOT
GND VCC SCL SDA
Soil Moisture Sensor

Transceiver



Working Flow



Working Flow

NODE

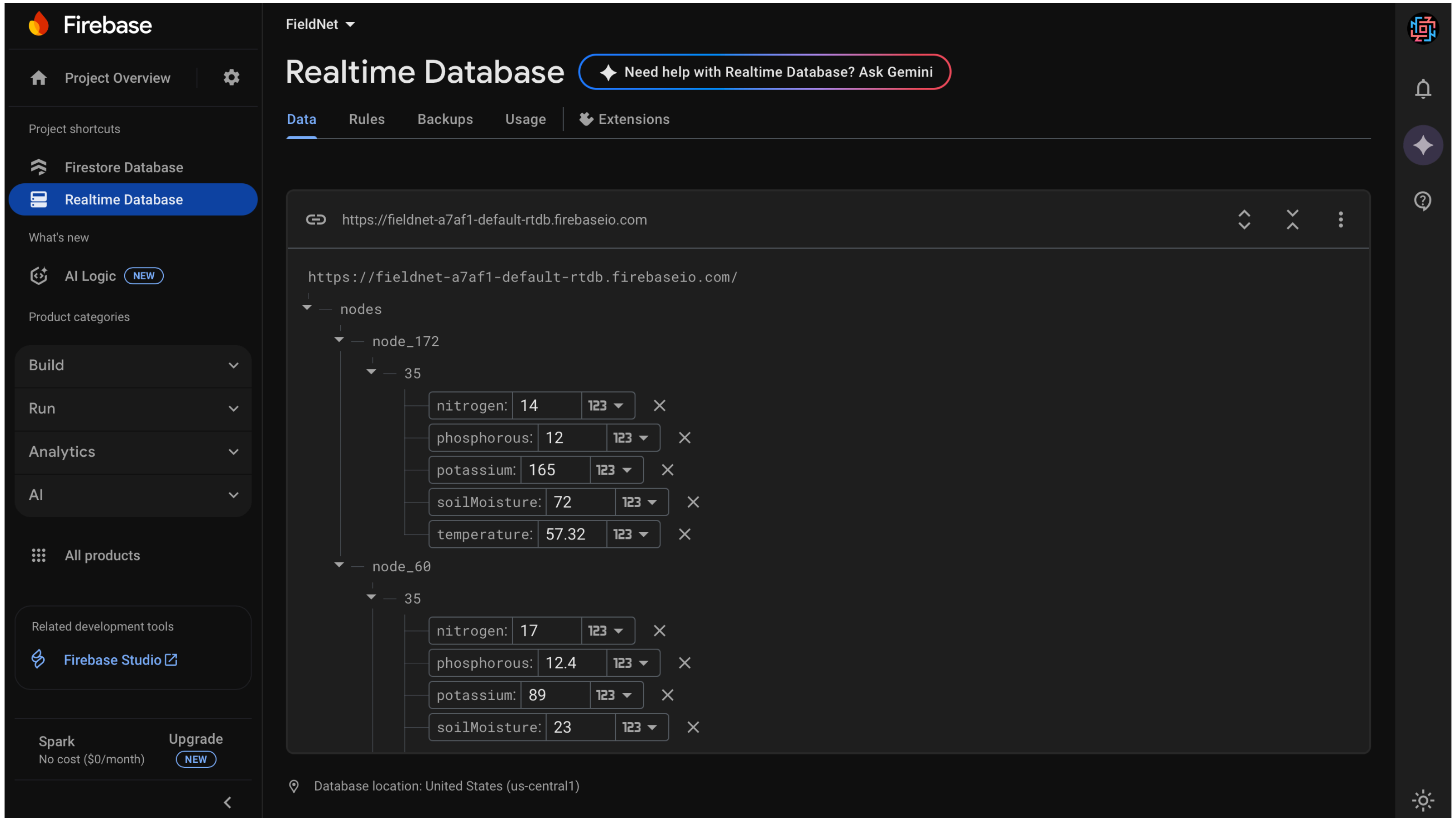
Start
↓
Read Temperature & Humidity Sensor
↓
Read Moisture & pH Neutron Sensor
↓
Process Sensor Data in RISC Controller
↓
Send Data via LoRa Transceiver
↓
Wait for Control Command via LoRa
↓
Continue Monitoring Sensors

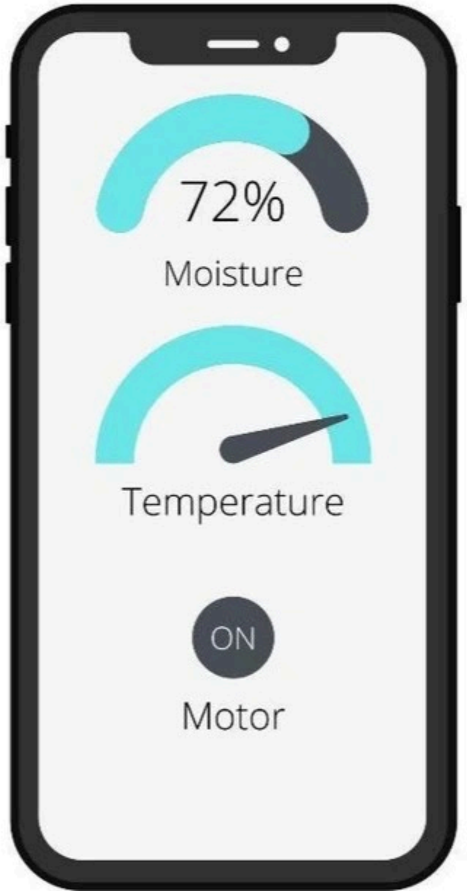
Activate Control Medium
↓
Repeat Cycle

Transceiver

Sensors Date
↓
LoRa Transceiver
↓
Gateway (RISC Controller + Display)
↓
Firebase Database
↙ ↘
User App Command Listener
↓ ↓
User Sends Cmd Gateway Receives Cmd
↓ ↓
Firebase Database
↓ ↓
Gateway Reads Cmd
↓ ↓
Actuator (Relay/Control Medium)

Dashboard


Firebase
Project Overview
Project shortcuts
Firestore Database
Realtime Database
What's new
AI Logic NEW
Product categories
Build
Run
Analytics
AI
All products
Related development tools
Firebase Studio
Spark
No cost ($0/month)
Upgrade
NEW
FieldNet
Realtime Database
Need help with Realtime Database? Ask Gemini
Data
Rules
Backups
Usage
Extensions
https://fieldnet-a7af1-default-rtdb.firebaseio.com
https://fieldnet-a7af1-default-rtdb.firebaseio.com/
nodes
node_172
35
nitrogen: 14
phosphorous: 12
potassium: 165
soilMoisture: 72
temperature: 57.32
node_60
35
nitrogen: 17
phosphorous: 12.4
potassium: 89
soilMoisture: 23
Database location: United States (us-central1)


72%
Moisture
Temperature
ON
Motor

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⚡ Results

You've set up a system on a one-acre farm. To monitor it, you've placed LoRa transmitters and receivers at opposite ends of the property.

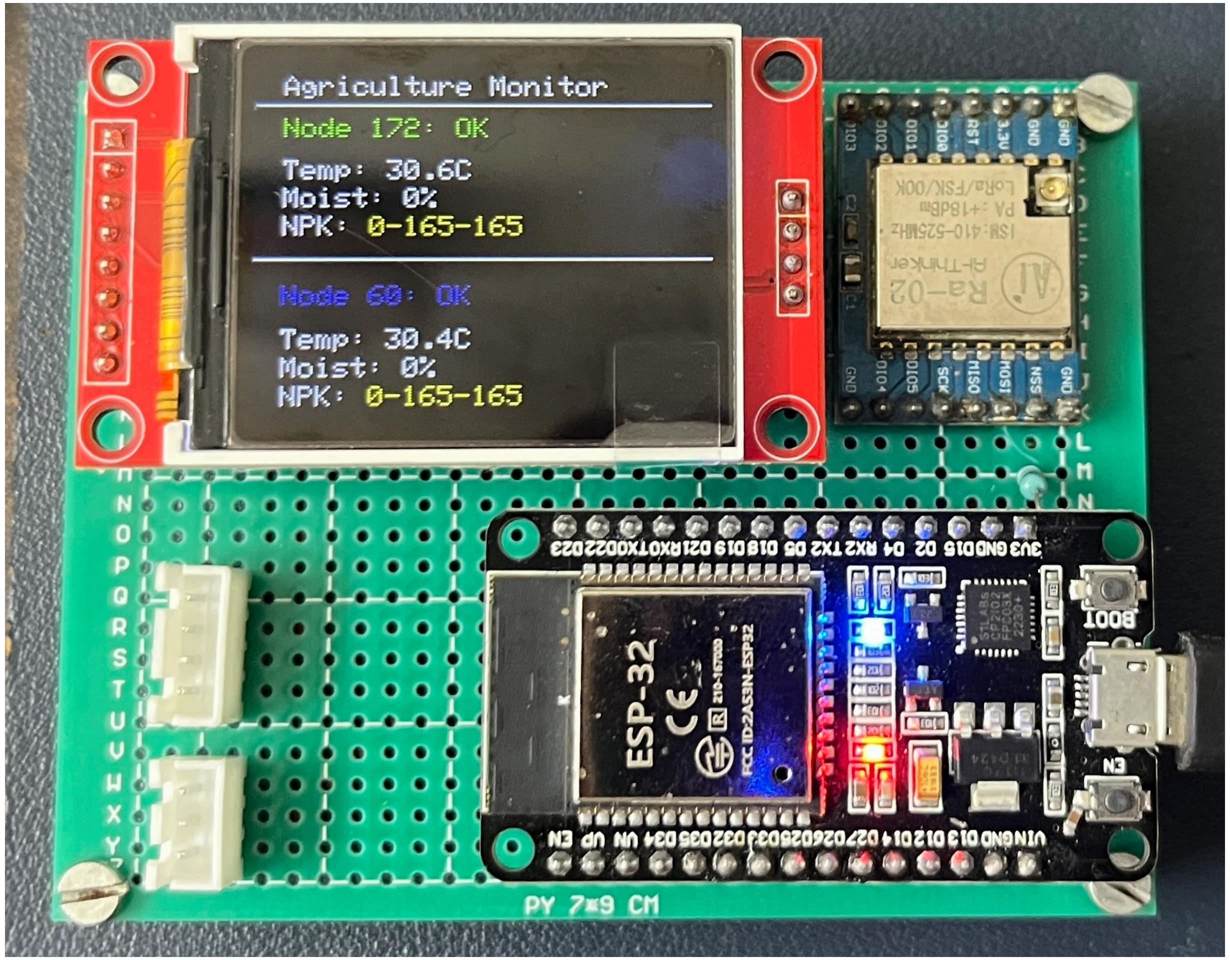
The farm itself has been divided into a grid of squares, each measuring 45 feet by 45 feet. I took signal strength measurements (called RSSI) at 20 different points within these grids.

The typical signal strength you observed across these locations was -137 dBm.

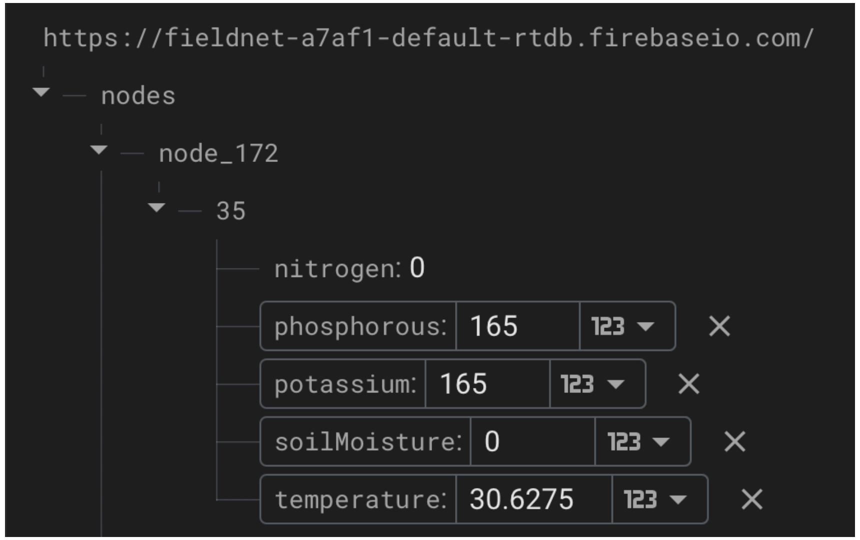
Benefits

1. Increased Efficiency Reduced waste of water and nutrients.
2. Better Yields Healthier plants, optimized growth conditions.
3. Reduced Labor Automated monitoring and control.
4. Data-Driven Decisions Farmers gain actionable insights.

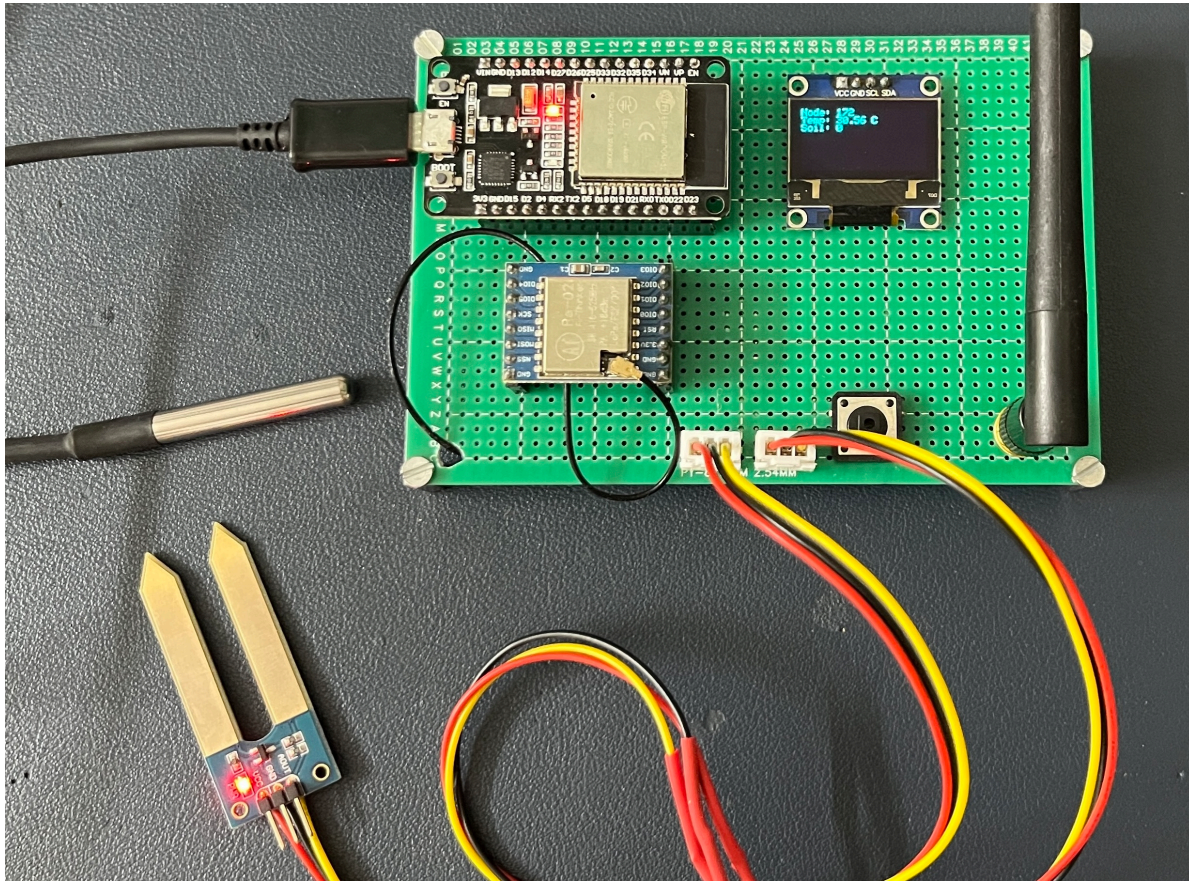
⚡ Results



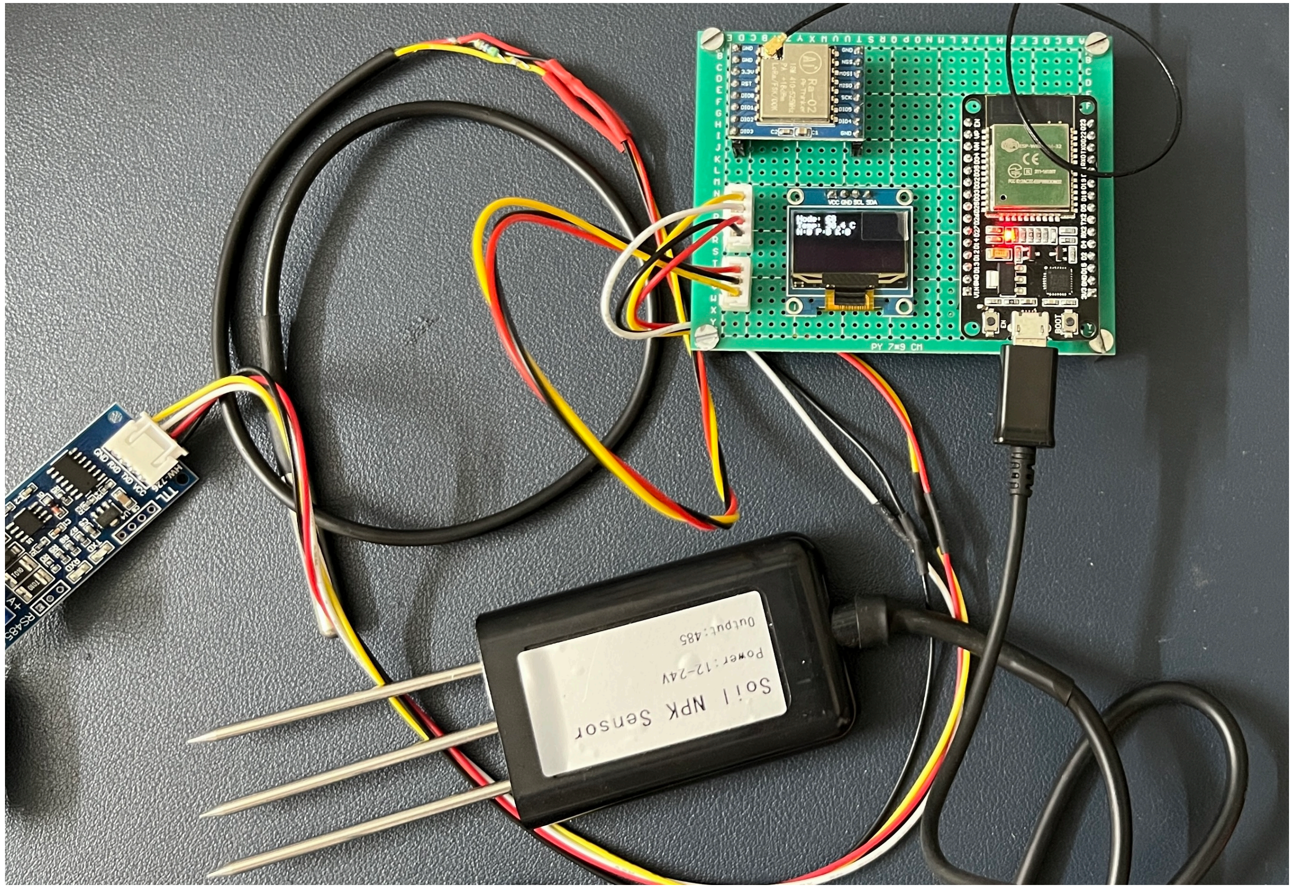
Transreceiver



Results

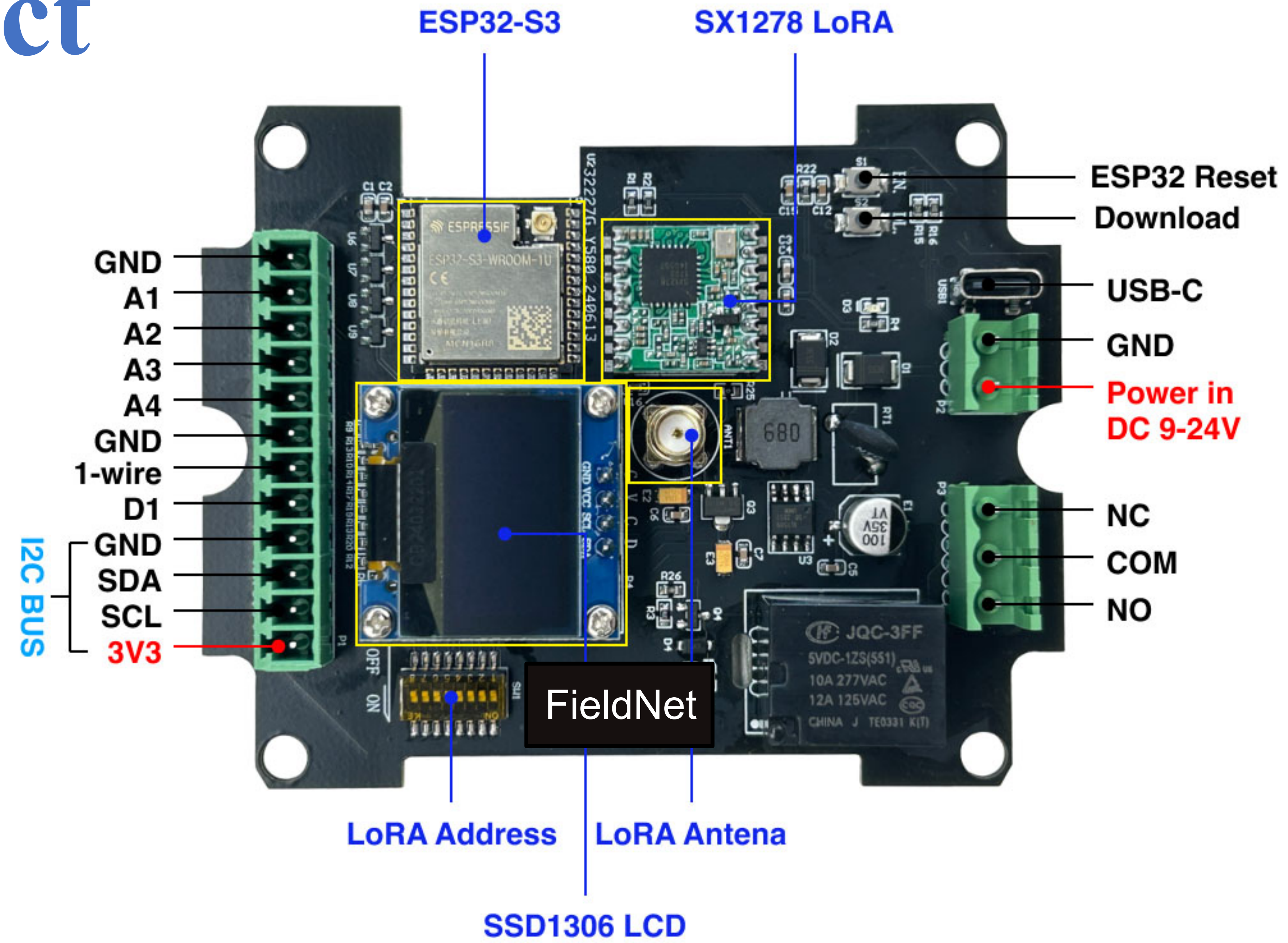


Node 1

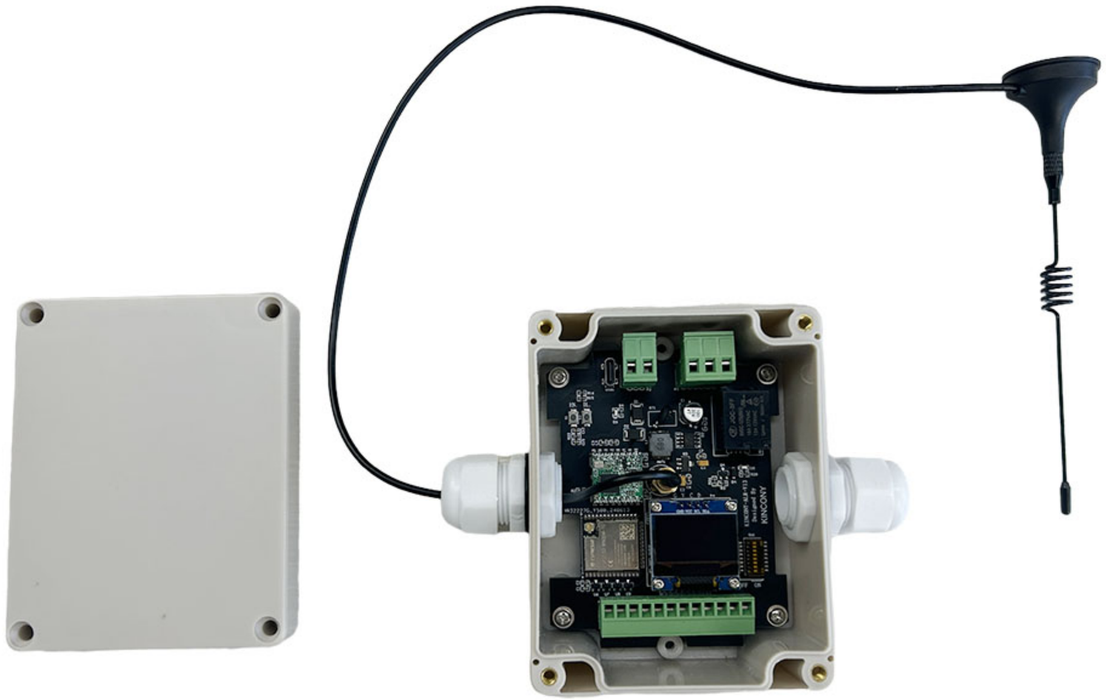


Node 2

Product


ESP32-S3
SX1278 LoRA
ESP32 Reset
Download
USB-C
GND
Power in
DC 9-24V
NC
COM
NO
GND
A1
A2
A3
A4
GND
1-wire
D1
GND
SDA
SCL
3V3
I2C BUS
FieldNet
LoRA Address
LoRA Antena
SSD1306 LCD

Product



Hardware Resource

ESP32-S3-WROOM-1U N16R8
SX1278 LoRa module:- Use by 433MHz
SSD1306 i2c OLED Display
Power supply:- DC9-24V
USB-C:- Download firmware
Buttons:- ESP32 Reset and Download
Relay:- MAX Load AC-250v 10A
Modbus RS485:- 1 Channel
1-wire:- 1 Channel
Digital input:- 1 Channel
I2C bus:- 1
Analog Input:- 4 channel dc 0-5v signal

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Conclusion

- Our smart farm system is a game-changer! By continuously monitoring conditions, intelligently processing data, and communicating wirelessly via LoRa, we can precisely control various farm operations.

- LoRa is the first wireless system that operates at km distances while consuming less than **10 μW**

- Inexpensive, flexible form factor connectivity solutions which can be integrated into everyday objects

- Feasible for wide-area deployment whole-home coverage, precision agriculture, long-range medical implants, and more

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Thank You!!

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Github: https://github.com/PrateekSinghRajput/Post-Graduate-Project-Details-2025